Article

Theory-Based Interpretability in Deep Knowledge Tracing via Grounded Transformers

Concepcion Labra [1], Olga C. Santos [2,*]

[1] Department of Artificial Intelligence, Computer Science School, UNED, 28040 Madrid, Spain; clabra@dia.uned.es
[2] PhyUM Research Center, Department of Artificial Intelligence, Computer Science School, UNED, 28040 Madrid, Spain; ocsantos@dia.uned.es
* Author to whom correspondence should be addressed.

Abstract

Knowledge Tracing (KT), which estimates how students' knowledge evolves during interactions with educational content, is a cornerstone of intelligent tutoring systems. While deep learning models achieve superior predictive performance, they lack interpretability, which is problematic in educational contexts where transparency is essential. We introduce *gTransformer*, a new type of grounded Transformer models bridging deep learning performance with intrinsic interpretability through *Representational Grounding*. The architecture adds theory-based parameters to student interaction sequences and uses attention mechanisms to transform these inputs into latent representations. These are projected into parameters that incorporate longitudinal context while preserving theoretical semantics to drive predictions through interpretable logic. Validation demonstrates: (1) structural encoding around theoretical constructs (probing selectivity $\Delta R^2 > 0.5$); (2) semantic alignment preserving pedagogical meaning ($\rho_T = 0.604$); and (3) functional alignment with quantified confidence. Interpretable predictions surpass the reference model (+17.0% AUC) with minimal interpretability cost versus black-box architectures. Critically, gTransformer enables context-aware personalization by differentiating students based on longitudinal learning trajectories rather than immediate responses, capturing patterns that traditional Markovian models cannot represent. This offers a practical path toward AI-driven personalization that is both accurate and interpretable, supporting educators in tailoring adaptive instruction.

Keywords: deep knowledge tracing; theory-based interpretability; grounded transformers; bayesian knowledge tracing; educational data analysis; personalized learning

1. Introduction

Knowledge Tracing (KT) [1] is a foundational problem in Intelligent Tutoring Systems and Learning Analytics [2]. It is formally defined as the supervised sequence modeling task of estimating a student's evolving mastery based on their historical interactions. The input to the system consists of a temporal sequence $\mathcal{S} = \{(q_1, r_1), (q_2, r_2), \ldots, (q_{t-1}, r_{t-1})\}$, where each tuple represents a question (or skill) q_i and the student's binary response $r_i \in \{0, 1\}$. The task objective is to predict the output $\hat{y}_t = P(r_t = 1 | q_t, \mathcal{S})$, signifying the probability of a correct response to the current question q_t given the observed history. By monitoring these longitudinal estimations, we can track the dynamic evolution of student knowledge, providing a data-driven foundation for personalized instruction and targeted pedagogical

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interventions. In an era where educational environments are becoming increasingly digital and diverse, the ability to accurately track and interpret student mastery is not merely a technical optimization but a critical requirement for effective, scalable, and learner-centred education.

Historically, the development of KT has been characterized by a dichotomy between two distinct paradigms [3]. The first, which predominated in the field for years, is characterized by probabilistic models such as Bayesian Knowledge Tracing (BKT) [1] and Factor Analysis Models based on logistic regression, including Additive Factor Models [4], Performance Factor Analysis [5], and Knowledge Tracing Machines [6]. Factor Analysis Models are theoretically supported by the Item Response Theory (IRT) [7], which has played a large role in educational assessment and measurement. These traditional approaches are intrinsically interpretable: they model learning as a process governed by explicit parameters related to pedagogical, cognitive or psychometric principles. Because the model structure mirrors a theoretical understanding of human learning, educators can inspect these values to validate the model's reasoning and trust its outputs. However, this transparency comes at the cost of representational rigidity; the simplifying assumptions required by these models often limit their predictive power, causing them to struggle with the complex, non-linear, and long-range dependencies inherent in real-world student behavior [8].

The second paradigm emerged with the advent of Deep Knowledge Tracing (DKT) [9], which leverages deep learning techniques ranging from initial Recurrent Neural Networks to more recent Transformer architectures [10]. The latter, based on encoder-decoder components and attention mechanisms, are especially relevant as they provide the foundation for current Large Language Models (LLMs). By treating student interaction histories as temporal sequences, Transformer models apply techniques similar to those that characterize LLMs. DKT models have achieved state-of-the-art predictive performance, significantly outperforming classical baselines due to a structural high capacity that allow them to extract latent features from vast amounts of data [3]. However, this substantial gain in predictive performance has come at the cost of model transparency, creating a significant interpretability gap. Deep learning models are notoriously opaque "black boxes", where the internal high-dimensional representations bear no direct correspondence to constructs that are interpretable in terms of human-understandable concepts or established domain knowledge. In a high-stakes domain such as education, where algorithmic decisions influence learning trajectories, grades, and opportunities, this lack of transparency raises critical concerns regarding accountability and trust [11].

Extensive research has sought to bridge this gap, with Bai et al. [12] classifying modalities into *post-hoc* and *ante-hoc* categories. While *post-hoc* methods aim to extract explanations from black-box models after training, *ante-hoc* approaches seek to achieve intrinsic interpretability by design, ensuring the model's internal logic is transparent from the outset. Within this latter category, recent models have attempted to achieve intrinsic transparency by integrating attention mechanisms [13,14] or by incorporating side information such as IRT parameters [15–17]. These efforts aim to make the decision-making process visible, moving away from pure black-box architectures toward systems whose internal states can be inspected.

However, from the perspective of domain experts such as teachers and pedagogical designers, these solutions frequently fall short. Post-hoc explanations often rely on technical relevance scores or local approximations that remain disconnected from established pedagogical principles [18] while most current ante-hoc methods do not produce outputs that map directly to human-understandable educational constructs. As noted in various surveys [12,19], when deep learning models incorporate domain knowledge, the primary objective is typically the optimization of predictive performance, while interpretability is often overlooked or left as an unaddressed concern.

Consequently, there is a critical need for new approaches that move beyond technical transparency toward true pedagogical interpretability. For a KT system to be confidently adopted in high-stakes educational environments, its estimations must be grounded in trusted and empirically validated principles derived from relevant fields such as pedagogy, cognitive science, or psychometrics. Educators require models that speak their language, tracking mastery not through opaque latent features, but through concepts explicitly defined by established frameworks, including the knowledge concepts, prior knowledge, or learning rates postulated by traditional approaches such as BKT and Factor Analysis Models.

To address this challenge, we introduce the *grounded Transformer (gTransformer)*, a variant of attention-based Transformers that implements a novel Representational Grounding methodology to bridge the gap between deep learning expressiveness and theoretical alignment. This approach explicitly anchors the Transformer's latent representations to the semantic constructs of an intrinsically interpretable reference model. The gTransformer architecture utilizes an encoder-decoder design with attention mechanisms to create context vectors that encode complex student interaction patterns. These vectors are projected into context-aware parameters, which then drive predictions using the reference model's interpretable logic. By doing so, the model's estimations can be explained and traced back to the foundational pedagogical principles and theoretical frameworks that govern the system's predictive logic. This architecture achieves intrinsic interpretability by providing "grounded interpretability," as estimations are conveyed through the same well-validated concepts employed in traditional interpretable systems.

Our approach contributes to bridging the dichotomy between high-capacity deep learning and domain-specific theoretical transparency, providing two practical advantages. First, compared to non-interpretable DKT models, gTransformer offers intrinsic pedagogical transparency by generating actionable parameters, such as prior knowledge and learning rates, that are directly meaningful to educators. Second, compared to traditional approaches, gTransformer is context-aware, i.e it is able to encode individual patterns taking into account the history of student interactions, overcoming the limitations of rigid population-level parameters and Markovian assumptions characteristic of BKT. For instance, it can distinguish between a learner requiring foundational remediation and one exhibiting high acquisition momentum even when their immediate local performance appears indistinguishable. By integrating these capabilities, gTransformer provides a foundation for truly individualized student-centered instruction, allowing educators to move beyond predicting *if* a student will succeed toward understanding *why* they are struggling.

Our research is guided by three primary research questions:

- RQ1: **Theory-Based Interpretability Through Grounded Transformers**
  Can DKT models achieve state-of-the-art predictive performance while providing interpretability grounded in established principles and theories? Specifically, can we design a Transformer architecture that produces pedagogically meaningful mastery estimations that are explainable through a causal and interpretable logic, such as BKT?
- RQ2: **Trade-Offs Between Predictive Performance and Interpretability**
  How do the metrics of the supervised, interpretable, and BKT predictions compare? What is the cost of interpretability in terms of AUC? How much predictive gain do the interpretable grounded predictions achieve compared to traditional BKT?
- RQ3: **Practical Value for Student-Centered Personalization**

How can the longitudinal context captured by a Transformer-based model enhance student-centered personalization, and how does this compare with the performance of population-based models like BKT?

The evaluation is designed to address each of these questions. First, to validate *theory-based interpretability* (RQ1), we employ a triple validation protocol consisting of: (1) *structural alignment* via diagnostic probing, confirming that Bayesian constructs (P_{L_0}, P_T) are the dominant organizing principles in the latent space (high selectivity $\Delta R^2 > 0.5$); (2) *semantic alignment*, which measures the monotonic consistency between grounded parameters and original theoretical priors to verify pedagogical integrity ($\rho_T = 0.604$); and (3) *functional alignment*, quantifying the confidence with which interpretable estimations can substitute for the supervised baseline. Second, we quantify the *accuracy–interpretability trade-off* (RQ2) across five benchmark datasets, demonstrating that interpretable predictions consistently surpass classical BKT (averaging +17.0% AUC improvement) with a low interpretability cost relative to black-box architectures. Finally, we demonstrate *student-centered personalization* (RQ3) through context-aware diagnostics. Using case study visualizations of learning trajectories, we illustrate how gTransformer differentiates students based on longitudinal learning history rather than immediate response patterns. By leveraging attention mechanisms to encode longitudinal context that traditional models fundamentally overlook, gTransformer establishes a robust foundation for individualized instruction. This context-driven approach enables educators to move beyond simple outcome prediction toward a understanding of how a student is learning over time.

The remainder of this paper is structured as follows. Section 2 reviews related work, while Section 3 describes the gTransformer model, detailing the construction of embeddings, the grounding mechanism, and the prediction logic. Section 4 presents the experimental setup, followed by a discussion of the empirical results in Section 5. Finally, Section 6 provides the conclusions.

2. Related Work

This section provides a review of research related to gTransformer. We first examine the theoretical foundations and limitations of BKT, which serves as our reference model for pedagogical grounding. We then trace the evolution of DKT from Recurrent Neural Networks to recent attention-based architectures. Following this, we categorize existing efforts to bridge the interpretability gap in DKT, contrasting post-hoc and ante-hoc methodologies. Finally, our model is mapped to taxonomic dimensions of Informed Machine Learning, establishing the connection with this paradigm.

2.1. Bayesian Knowledge Tracing

For this study, we selected BKT [1] as our reference theoretical framework. Student action sequences in Intelligent Tutoring Systems can be modeled as Hidden Markov Models, where the probability of a transition depends only on the previous state [20]. BKT models the learning process as a Hidden Markov process governed by four parameters that provide a causal narrative of student learning progress, which is intuitive to educators and aligned with cognitive science principles:

- **Initial Mastery**: The probability a student knows a skill prior to practice
- **Learn Rate**: The probability of transitioning from unlearned to learned states after a practice opportunity
- **Guess**: The probability of a correct response despite a lack of mastery
- **Slip**: The probability of an incorrect response despite mastery

Standard BKT suffers from a significant limitation: it typically operates at a population level, estimating a single set of parameters for all students interacting with a given skill. While *Individualized BKT* approaches exist, they are often computationally expensive and struggle with data sparsity [21]. The Representational Grounding used in the gTransformer model offers a new way to address individualization by leveraging the capabilities of Transformers to compress student interaction history into a context vector. This allows the model to infer personalized initial knowledge and learning rates dynamically, transforming population-level BKT insights into high-granularity student diagnostics.

2.2. Deep Knowledge Tracing

The application of deep learning to KT was initiated by the DKT model [9], which framed student performance as a sequence modeling task. While pioneering, DKT was constrained by the inherent limitations of Recurrent Neural Networks (RNNs). These architectures are difficult to train [22] and their sequential nature precludes parallelization within training examples, a bottleneck that becomes critical at longer sequence lengths where memory constraints limit efficiency.

The introduction of attention-based architectures [10], such as Self-Attentive Knowledge Tracing (SAKT) [13] and Separated Self-AttentIve Neural Knowledge Tracing (SAINT) [23], addressed these limitations by enabling parallel processing of interaction histories. This marked a significant shift toward the attention-based Transformer architectures that characterize modern Large Language Models (LLMs). By replacing recurrent units with multi-head attention, these models capture long-range dependencies more effectively, allowing for a non-Markovian representation of a student's entire history. More recently, Attentive Knowledge Tracing (AKT) [14] further refined this approach by incorporating skill difficulty and context-aware attention mechanisms that weigh past interactions based on their temporal distance.

Beyond the prominent recurrent and attention-based paradigms, the research community has explored several other architectural directions to capture the multifaceted nature of student learning. Notable categories include Memory-Augmented models that utilize external storage to explicitly track concept mastery, Graph-Based approaches that model the topological relationships between skills, and specialized variations such as Text-Aware or Forgetting-Aware models that incorporate exercise semantics and temporal decay into the tracing process [3]. While these variants exploit diverse dimensions of domain knowledge to augment predictive performance, achieving intrinsic interpretability is generally not prioritized as a primary architectural objective.

2.3. Interpretability in Deep Knowledge Tracing

Interpretability in DKT remains an open challenge despite a variety of attempts, which can be categorized into two primary modalities: *post-hoc* and *ante-hoc*. Post-hoc methods attempt to explain trained black-box models by utilizing either model-specific techniques, such as Layer-Wise Relevance Propagation (LRP) [24], or model-agnostic approaches like Local Interpretable Model-agnostic Explanations (LIME) [25]. However, Rudin [18] argues that relying on post-hoc explanations for high-stakes decision-making is fundamentally flawed. For teachers and domain experts, these methods offer limited utility as they produce explanations tied to technical model structures rather than pedagogically sound principles, necessitating significant technical expertise for meaningful interpretation.

The second category, *ante-hoc* methods, pursues intrinsic interpretability through two primary approaches. The first consists of traditional models that rely on transparent structures, such as BKT or Factor Analysis Models. The second encompasses DKT variants that achieve interpretability by incorporating additional modules or utilizing side information [12]. Among these, attention mechanisms have emerged as the most prominent architectural extensions. They have been integrated into a wide range of models, from the pioneering SAKT [13] to more recent architectures like AKT [14], which achieves high predictive performance by utilizing multi-scale attention to capture interaction patterns across varying temporal resolutions. However, relying solely on attention weights presents similar practical obstacles to those of post-hoc methods for educational practitioners, as these mechanisms do not provide an explicit interpretation grounded in pedagogically sound principles.

In the integration of side information for ante-hoc interpretability, IRT [7] has served as the most prominent foundation. It estimates the probability of a correct response based on a student's latent ability and the item's difficulty. This theory has been coupled with DKT models such as Deep-IRT [15], which utilizes dynamic key-value memory networks to capture learners' trajectories and uses inferred abilities and difficulties to predict performance. Similarly, TC-MIRT [16] embeds multidimensional IRT parameters to predict student states and generate interpretable parameters. Other relevant models include KIKT [26], which harnesses IRT to simulate student performance; LANA [27], which adopts an IRT variant for ability clustering; and QIKT [17], which features question-centric representations integrated with an interpretable IRT layer. Beyond IRT, researchers have leveraged diverse principles, such as constructive learning [28,29], learning and forgetting curves [30], finite state automata (FSAs) [31], classical test theory (CTT) [32], and monotonicity constraints [33].

In contrast to these models, whose diagnostic utility is often restricted by the specificity and scope of the single underlying theory serving as side information [12], gTransformer offers a more flexible framework. It can be grounded in any established theoretical reference, effectively decoupling the expressiveness of deep learning from the constraints of specific pedagogical models.

2.4. Informed Machine Learning

The integration of domain knowledge into deep learning has been broadly explored through conceptual frameworks such as Theory-Guided Data Science (TGDS) [34], Physics-Informed Machine Learning (PIML) [35], and Informed Machine Learning (IML) [19].

TGDS emerged as a paradigm to address the limitations of purely data-driven "black-box" models, which may produce results inconsistent with established domain knowledge [34]. By explicitly requiring scientific consistency, TGDS integrates theoretical insights into the learning process to ensure that models remain scientifically plausible. This integration is typically achieved via theory-guided learning where domain-specific invariants serve as regularization terms, or through the design of neural architectures that respect the structural constraints of the problem [34,36]. A prominent implementation of this paradigm is Physics-Informed Neural Networks (PINNs) [35,37], which embed differential equations directly into the loss function, forcing the model to satisfy theoretical constraints while learning from empirical data.

A widely adopted technique to enforce this consistency involves incorporating domain-specific constraints into the loss function as follows [34,38]:

$$\mathcal{L} = \mathcal{L}_{TRN}(Y_{true}, Y_{pred}) + \gamma \mathcal{L}_{THEORY}(Y_{pred}), \tag{1}$$

where $\mathcal{L}_{TRN}$ measures the supervised error between true labels Y_{true} and predicted labels Y_{pred}, while $\mathcal{L}_{THEORY}$ penalizes violations of theoretical constraints, weighted by the hyperparameter γ.

Von Rueden et al. [19] formalized these efforts into an IML taxonomy, which categorizes knowledge integration by its source, representation, and the stage of the machine learning pipeline where it is incorporated. Knowledge can be represented as algebraic equations, logical rules, or probabilistic priors, and integrated at various stages: within the training data, through the design of tailored architectures, or by influencing the model's final output.

We operationalized these taxonomic dimensions in gTransformer by integrating theoretical priors from an interpretable BKT model into the inputs and employing a multi-objective loss function to anchor the model's predictions to BKT parameters. However, while most IML approaches focus on using prior knowledge to maximize predictive accuracy [19], gTransformer aims to achieve theory-based interpretability while offering competitive predictive performance, as detailed in Section 3.1.

3. The gTransformer Model

This section details the design and implementation of gTransformer, a hybrid architecture that bridges the gap between deep learning expressiveness and theoretical alignment by grounding the Transformer's latent representations in concepts established by a reference model. We begin by mapping its components to the taxonomic dimensions of IML, clarifying how theoretical priors and empirical data are processed using specialized inductive biases and multi-objective optimization. We then provide a comprehensive description of the model's architecture, detailing its functional layers: from embeddings enriched with prior knowledge to the core attention-based encoder-decoder engine. We explain the projection mechanism enabling the estimation of individualized, pedagogically meaningful parameters, which are fed into an output head implementing BKT logic to produce interpretable predictions. Finally, we detail the loss functions that anchor the model's predictions to BKT parameters.

3.1. *Dimensions of Informed Machine Learning in gTransformer*

Figure 1 illustrates the information flow through gTransformer, mapped to the taxonomic dimensions of IML as defined by Von Rueden et al. [19].

Within the context of our architecture, the operationalization of these dimensions can be explained as follows:

- **Prior Knowledge**: The source of theoretical information comprises simulation results from a reference model, typically expressed as parameter values that represent foundational pedagogical constructs.
- **Training Data**: Empirical student interaction histories are augmented with prior knowledge to train the model.
- **Hypothesis Set**: An attention-based Transformer architecture is employed to provide the inductive biases necessary to lead the model toward aligned states.
- **Learning Algorithm**: The model is optimized via a multi-objective loss function formulated to balance the trade-off between predictive performance and pedagogical interpretability.
- **Final Hypothesis**: A set of learned weights that drive latent representations from which parameter values aligned with the reference model can be extracted.

3.2. *Architecture*

The gTransformer architecture, illustrated in Figure 2, implements a theory-guided deep learning framework that integrates BKT concepts throughout the processing pipeline. The architecture is organized into six functional layers, which are detailed in the following subsections. Specifically, we: (i) begin with input data comprising student interaction sequences along with prior knowledge from the Bayesian reference model (Section 3.2.1); (ii) proceed to difficulty-aware embeddings that encode skill-specific variations (Section 3.2.2);

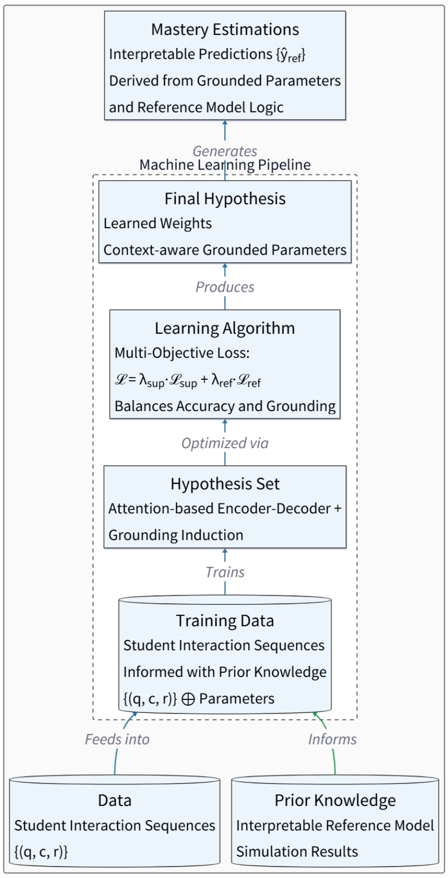

Figure 1. Dimensions of Informed Machine Learning in gTransformer.

(iii) describe the attention-based encoder-decoder mechanism used to construct contextualized latent representations (Section 3.2.3); (iv) explain how grounded parameters are obtained combining theoretical bases with contextual projections (Section 3.2.4); (v) present the three parallel output heads intended to generate supervised predictions, interpretable predictions, and representation probes (Section 3.2.5); and (vi) conclude with the multi-objective loss function that orchestrates the training process to achieve both predictive performance and theory-based interpretability (Section 3.2.6).

3.2.1. Input Components

The data flow begins with the ingestion of student interaction sequences enriched with population-level parameters from a pre-fit BKT reference model. Student interactions are represented as temporal sequences of tuples (q_t, c_t, r_t), where q_t denotes the question identifier, c_t represents the associated knowledge component, concept or skill, and $r_t \in \{0, 1\}$ indicates the binary correctness of the student's response at timestep t.

These empirical interaction traces are augmented with theoretical priors extracted from the BKT reference model. For each knowledge component c, we obtain population-level estimates of four fundamental BKT parameters: prior knowledge $P(L_0)$ (the probability that a randomly selected student has already mastered concept c before any practice), learning rate $P(T)$ (the probability of transitioning from non-mastery to mastery after a single learning opportunity), guess probability $P(G)$ (the likelihood of answering correctly despite not having mastered the concept), and slip probability $P(S)$ (the likelihood of answering incorrectly despite having mastered the concept). For the sake of simplicity, only $P(L_0)$ and $P(T)$ undergo the grounding process. In contrast, we use directly the

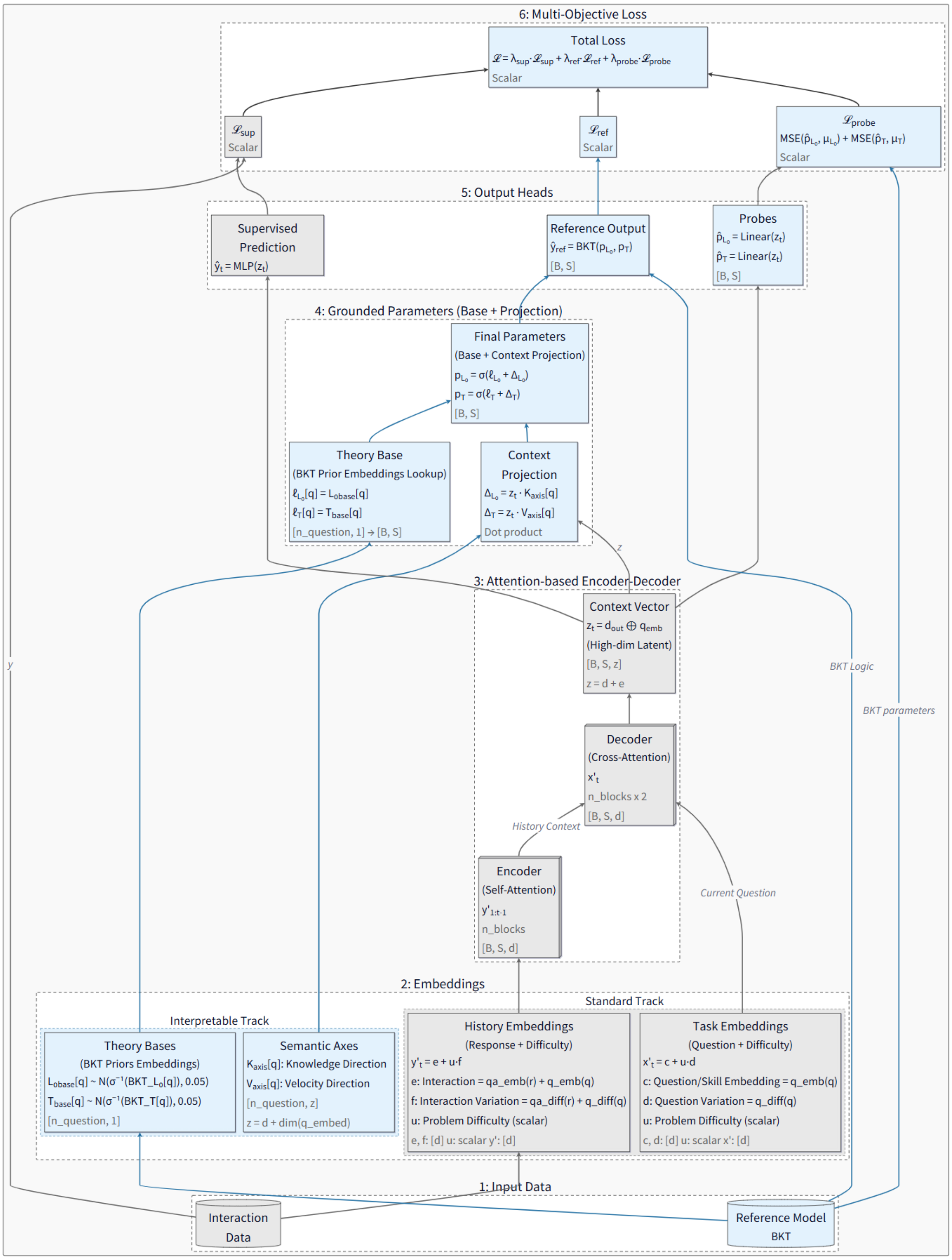

Figure 2. Functional layers of the gTransformer architecture. Gray components denote standard Transformer elements while blue components denote gTransformer-specific elements.

values estimated by BKT for the parameters $P(G)$ and $P(S)$ to calculate the interpretable predictions $\hat{y}_{\text{ref}}$.

3.2.2. Prior-Enriched Embeddings

For question embeddings, we define:

$$x'_t = c_q + u_q \cdot d_c \tag{2}$$

where $c_q \in \mathbb{R}^d$ represents the invariant concept base embedding (encoding the semantic identity of the knowledge component), $u_q \in \mathbb{R}$ is a scalar difficulty parameter specific to question q, and $d_c \in \mathbb{R}^d$ is the concept-specific difficulty variation axis. This formulation allows multiple questions targeting the same concept to share a common semantic core (c_q) while differing in their position along a learned difficulty direction (d_c), with the magnitude of displacement controlled by the per-question difficulty scalar (u_q).

Similarly, for interaction history embeddings (encoding both the question and the student's response), we employ:

$$y'_t = e_{c,r} + u_q \cdot f_{c,r} \tag{3}$$

where $e_{c,r} \in \mathbb{R}^d$ is the interaction base embedding (jointly encoding the concept c and response outcome r), and $f_{c,r} \in \mathbb{R}^d$ is the interaction-specific difficulty variation axis.

These embeddings serve as the primary inputs to the Transformer's attention mechanism, providing multi-dimensional representations that enable attention-driven discovery of relevant behavioral patterns across the student's learning history.

3.2.3. Attention-Based Encoder-Decoder

The core processing engine of gTransformer is a stacked Transformer architecture composed of n encoder blocks followed by $2n$ decoder blocks. The encoder processes the student's longitudinal interaction history through n sequential self-attention layers, transforming the sequence of interaction embeddings $\{y'_1, y'_2, \ldots, y'_{t-1}\}$ into a tensor of contextualized latent representations $Y_{\text{enc}} \in \mathbb{R}^{(t-1)\times d}$. Each encoder block applies multi-head self-attention with monotonic causal masking to prevent information leakage from future timesteps, followed by feed-forward networks and residual connections with layer normalization, following the standard Transformer architecture [10].

The decoder architecture employs an alternating pattern of $2n$ sequential layers. Odd-numbered decoder layers (layers $1, 3, 5, \ldots, 2n-1$) perform self-attention over the current question embeddings x'_t, allowing the model to examine multiple aspects of the current task without incorporating response history. Even-numbered decoder layers (layers 2, 4, 6, $\ldots$, $2n$) perform cross-attention between the current question representation and the encoder's output Y_{enc}, retrieving relevant knowledge from the student's encoded learning trajectory.

The final decoder layer produces an output tensor $d_{\text{out}} \in \mathbb{R}^{t\times d}$, which is concatenated with the question embedding to form the high-dimensional context vector:

$$z_t = [d_{\text{out},t}; x'_t] \in \mathbb{R}^z \tag{4}$$

where $z = d + e$ represents the combined dimensionality of the decoder output and question embedding spaces. This context vector serves as a comprehensive latent encoding of the student's current cognitive state, synthesizing information from both the observed interaction history and the current task characteristics.

3.2.4. Grounded Parameters

To ensure pedagogical interpretability, gTransformer estimates BKT parameters through an additive composition mechanism operating in logit space. For each knowledge component c, we define theoretical base parameters $\ell_{L_0,c}$ and $\ell_{T,c}$, initialized as the logit-transformed population-level BKT priors:

$$\ell_{L_0,c} = \text{logit}(P(L_0)_c), \quad \ell_{T,c} = \text{logit}(P(T)_c) \tag{5}$$

These bases are implemented as scalar embeddings with Gaussian initialization centered at the theoretical values.

Contextual adjustments are computed via dot products between the latent context vector z_t and learnable semantic axes. For each concept c, we define a knowledge axis $K_c \in \mathbb{R}^z$ and a learning velocity axis $V_c \in \mathbb{R}^z$. The contextual projections, representing the student-specific Knowledge Gap (k_s) and Learning Velocity (v_s), are calculated as:

$$k_{s,t} = z_t \cdot K_c, \quad v_{s,t} = z_t \cdot V_c \tag{6}$$

These dot products project the high-dimensional latent state onto interpretable semantic dimensions, quantifying how much the student's observed behavior deviates from the population average along the dimensions of prior mastery and learning rate.

The final grounded parameters are obtained through additive composition in logit space, followed by sigmoid activation to ensure valid probability bounds:

$$p_{L_0,t} = \sigma(\ell_{L_0,c} + k_{s,t}), \quad p_{T,t} = \sigma(\ell_{T,c} + v_{s,t}) \tag{7}$$

This design ensures that the model's parameter estimates remain structurally anchored to BKT theory, with the base terms providing pedagogically sound initialization, while contextual projections enable individualized adjustments driven by observed student behavior.

3.2.5. Output Heads

The gTransformer model generates predictions through three parallel output heads, each serving a distinct functional role in the architecture's dual objectives of predictive accuracy and pedagogical interpretability.

Supervised Prediction Head: A multi-layer perceptron (MLP) directly maps the context vector z_t to a correctness prediction $\hat{y}_t = \text{MLP}(z_t)$. This head is optimized purely for predictive accuracy via binary cross-entropy loss, leveraging the full expressive capacity of the Transformer's learned representations without theoretical constraints.

Interpretable Output Head: The grounded BKT parameters $(p_{L_0,t}, p_{T,t})$ are passed through an implementation of BKT logic, which performs a retrospective belief update walk over the student's interaction history. At each timestep, the BKT logic computes the probability of correctness $\hat{y}_{\text{ref},t}$ using the standard BKT update equations with the model's grounded parameters and the fixed population-level guess and slip values. These predictions are intrinsically interpretable because they can be directly tracked to established pedagogical constructs, providing a transparent causal narrative that justifies each estimation through the theoretical lens of BKT. This head ensures that the grounded parameters retain their intended semantic meaning by requiring them to produce valid predictions when subjected to the reference model's reasoning process.

Linear Probe Heads: Two dedicated linear layers extract BKT parameter estimates directly from the context vector: $\hat{p}_{L_0,t} = \sigma(W_{L_0} z_t)$ and $\hat{p}_{T,t} = \sigma(W_T z_t)$. Unlike the grounded parameter projections (which use concept-specific semantic axes), these probes are global linear extractors shared across all concepts. They serve to verify that the Transformer's internal representations explicitly encode the targeted pedagogical constructs in a linearly accessible manner.

3.2.6. Multi-Objective Loss

The training objective balances predictive performance with pedagogical grounding through a weighted combination of three loss terms:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{sup}}\mathcal{L}_{\text{sup}} + \lambda_{\text{ref}}\mathcal{L}_{\text{ref}} + \lambda_{\text{probe}}\mathcal{L}_{\text{probe}} \tag{8}$$

The supervised loss $\mathcal{L}_{\text{sup}}$ measures the binary cross-entropy between the MLP's direct predictions $\hat{y}_t$ and the ground truth responses r_t, driving the model to maximize predictive accuracy:

$$\mathcal{L}_{\text{sup}} = -\frac{1}{N}\sum_{t=1}^{N}[r_t \log(\hat{y}_t) + (1-r_t)\log(1-\hat{y}_t)] \tag{9}$$

The reference output loss $\mathcal{L}_{\text{ref}}$ applies the same binary cross-entropy metric to the BKT logic head's predictions $\hat{y}_{\text{ref},t}$, ensuring that the grounded parameters $(p_{L_0,t}, p_{T,t})$ produce pedagogically meaningful predictions when constrained to operate through BKT's causal inference mechanism:

$$\mathcal{L}_{\text{ref}} = -\frac{1}{N}\sum_{t=1}^{N}[r_t \log(\hat{y}_{\text{ref},t}) + (1-r_t)\log(1-\hat{y}_{\text{ref},t})] \tag{10}$$

The probing loss $\mathcal{L}_{\text{probe}}$ enforces global alignment between the Transformer's latent representations and BKT constructs by minimizing the mean squared error between linear probe predictions and population-level BKT targets:

$$\mathcal{L}_{\text{probe}} = \sum_{k\in\{L_0,T\}} \text{MSE}(\hat{p}_{k,t}, P(k)_c) \tag{11}$$

This loss implements an "active grounding" mechanism, forcing the context vector z_t to organize its internal structure such that pedagogical parameters can be extracted via simple linear transformations. By constraining the latent space globally rather than merely regularizing output parameters locally, the probing loss ensures that interpretability is structurally embedded throughout the model's representations.

The default configuration uses weight coefficients $\lambda_{\text{sup}} = 1.0$, $\lambda_{\text{ref}} = 0.5$, and $\lambda_{\text{probe}} = 1.0$, balancing supervised performance with theoretical grounding while prioritizing global latent space alignment through active probing.

4. Experimental Setup

This section details the experimental framework employed to evaluate gTransformer, covering training configurations, five benchmark educational datasets, and the baseline models selected for comparison.

4.1. Training Setup

The gTransformer model was implemented in `PyTorch` using the `pyKT` library [39] for standardized data preprocessing, dataset splitting, and baseline models implementation.

Training and evaluation followed a standard five-fold cross-validation protocol using an 80/20 train/test split. Predictive performance was measured using the Area Under the Curve (AUC), applying the question-level, mean late-fusion protocol as recommended in [39]. Hyperparameter configurations for baseline models were aligned with the optimized values reported in the official `pyKT` repository to ensure fair comparison. Parameters for gTransformer are detailed in Table 3 and Table 7. All experiments were conducted on a machine equipped with 40 CPU cores and 8 NVIDIA Tesla V100-SXM2 GPUs with 32 GB HBM2 each.

4.2. Datasets

We evaluated gTransformer using five benchmark datasets that are among the most frequently cited and utilized for knowledge tracing benchmarks [39].

- **ASSISTments 2009 (AS2009)**: Collected from the ASSISTments platform during the 2009-2010 school year, this dataset consists of math exercises and has served as the de facto standard benchmark for KT research for the past decade [40].
- **ASSISTments 2015 (AS2015)**: A larger release from the same platform (2015), featuring the highest volume of students among the ASSISTments datasets while focusing on a specific set of 100 knowledge concepts [40].
- **Algebra 2005 (AL2005)**: Released as part of the KDD Cup 2010 EDM Challenge, it contains detailed step-level responses from 13-14 year old students solving Algebra problems [41].
- **Bridge to Algebra 2006 (BDG2006)**: Also from the KDD Cup 2010 Challenge, this dataset focuses on the Bridge to Algebra tutoring system and features a significantly higher number of total interactions [41].
- **NeurIPS 2020 Education Challenge (NIPS34)**: A more recent diagnostic dataset from the Eedi platform containing student answers to multiple-choice math questions, characterized by high-quality subjects trees used for concept mapping [42].

The characteristics of the datasets are summarized in Table 1:

Dataset	Sequences (N_s)	Concepts (N_c)	Questions (N_q)	Interactions (N_i)
AS2009	4,217	123	26,688	346,860
AS2015	19,917	100	—	708,631
AL2005	574	112	210,710	809,694
BDG2006	1,146	493	207,856	3,679,199
NIPS34	4,918	57	948	1,382,727

Table 1. Statistics for the five benchmark datasets used in this study. N_s, N_c, N_q, and N_i denote the total number of sequences (students), knowledge components (concepts or skills), unique questions, and student-item interactions, respectively.

4.3. Baseline Models

For the comparative analysis, we selected six models that are among the most frequently mentioned baselines [39]: DKT [9], DKVMN [43], SAKT [44], SAINT [23], AKT [14], and ATKT [45]. These models represent a comprehensive cross-section of the recurrent (DKT), memory-augmented (DKVMN), adversarial (ATKT), and attention-based (SAKT, SAINT, AKT) paradigms, providing a robust and diverse benchmark for evaluating gTransformer.

5. Results and Discussion

5.1. Predictive Performance

To evaluate the predictive performance of gTransformer, we utilized the Area Under the Curve (AUC) across the five benchmark datasets described in Section 4.2. The experimental results are summarized in Table 2.

Comparing these results with baseline models, we observe that gTransformer achieves highly competitive performance across all benchmarks. For this comparison, we utilize a

Model	AS2009	AS2015	AL2005	BDG2006	NIPS34
AKT	0.7825	0.7081	**0.8306**	**0.8208**	**0.8033**
ATKT	0.7715	**0.7810**	0.7995	0.7889	0.7665
DKT	0.7528	0.7031	0.8149	0.8015	0.7689
DKVMN	0.7440	0.7013	0.8054	0.7983	0.7673
SAKT	0.7263	0.6947	0.7880	0.7740	0.7517
SAINT	0.6921	0.6836	0.7775	0.7781	0.7873
gTransformer	**0.7831**	0.7078	0.8240	0.8148	0.8006

Table 2. AUC across the datasets evaluated on questions level with mean late fusion.

version with interpretability mechanisms ablated ($\lambda_{\text{ref}} = 0$ and $\lambda_{\text{probe}} = 0$), functioning as a pure neural model to establish its maximum representational capacity. The hyperparameter configurations used for each dataset are shown in Table 3. On AS2009, this configuration achieves an AUC of 0.7831, outperforming all other models. It ranks second on AL2005, BDG2006, and NIPS34, surpassed only by AKT, and third on AS2015, behind ATKT and AKT. These results demonstrate gTransformer's flexibility in delivering top-tier predictive performance alongside theory-based interpretability, offering a robust and transparent alternative for high-stakes educational environments where pedagogical grounding is paramount.

Parameter	AS2009	AS2015	AL2005	BDG2006	NIPS34
Architecture Configuration					
Embedding dimension (d_{model})	64	64	64	64	64
Transformer blocks (n_{blocks})	4	4	4	4	4
Attention heads (num_attn_heads)	4	4	4	4	4
Feed-forward dimension (d_{ff})	256	256	256	256	256
Training Configuration					
Optimizer ($optimizer$)	Adam	Adam	Adam	Adam	Adam
Learning rate ($learning\_rate$)	10^{-4}	10^{-4}	10^{-4}	10^{-4}	3×10^{-4}
Batch size ($batch\_size$)	64	64	64	64	64
Dropout rate ($dropout$)	0.1	0.1	0.1	0.1	0.1
Epochs ($epochs$)	200	200	200	200	200
Loss Weights					
Supervised (λ_{sup})	1.0	1.0	1.0	1.0	1.0
Reference (λ_{ref})	0	0	0	0	0
Probing (λ_{probe})	0	0	0	0	0

Table 3. Parameter values used to train gTransformer across all datasets in ablated mode.

5.2. Theory-Based Interpretability Through Grounded Transformers (RQ1)

We will use the following hypotheses to validate the RQ1 research question posed in section 1:

- **H1.1: Structural Encoding**
  Latent representations in the gTransformer model are structurally organized around BKT constructs (initial mastery P_{L0} and learning rate P_T) as the dominant organizing principle.
- **H1.2: Semantic Alignment**
  The extracted parameters are semantically aligned with meaningful BKT priors.
- **H1.3: Functional Alignment**

The interpretable predictions derived from extracted parameters can be used with quantified confidence, enabling educators to identify when theory-grounded explanations are trustworthy versus when additional validation is recommended.

The objective of this triple validation is to demonstrate that grounding constraints actively shape the model's internal structure, not merely correlate with outputs, producing interpretable predictions. It addresses a fundamental challenge in educational AI: whether neural models can internalize pedagogical theory rather than learning arbitrary features that happen to predict well, and whether such interpretability can be achieved without sacrificing the representational power that makes deep learning effective.

5.2.1. Structural Alignment via Diagnostic Probing (H1.1)

To verify whether gTransformer's latent representations genuinely internalize pedagogical concepts, we employ *diagnostic probing with control tasks* [46,47]. This approach trains linear Ridge regression models to act as diagnostic probes, checking if the original BKT theoretical parameters (p_{L_0} and p_T) can be extracted from the final hidden states z_t produced by gTransformer.

Methodology: We measure *fidelity* as the quality of the parameter recovery from the latent representations, measured via R^2 and Pearson correlation r. To distinguish genuine structural encoding from spurious correlations, *selectivity* is calculated to compare probe performance on true BKT targets against a control task with randomly shuffled target values. If probes perform equally well on shuffled targets, the representations lack genuine structural encoding. Conversely, high selectivity ($\Delta R^2 > 0.5$) demonstrates that BKT constructs are the primary organizing axes rather than incidental patterns.

- **Fidelity**: The R^2 (coefficient of determination) is calculated on the test set, representing the proportion of the variance in the theoretical parameters that is predictable from the gTransformer's latent space:

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

  where y_i is the actual BKT theoretical value, $\hat{y}_i$ is the value predicted by the linear probe from the hidden state z_t, and $\bar{y}$ is the mean of the theoretical values.
- **Selectivity**: Compares true performance against a shuffled control:

$$\Delta R^2 = R^2_{\text{true}} - R^2_{\text{control}}$$

Results: Table 4 summarizes the diagnostic probing results across four datasets with varying structural properties. We observe strong evidence of BKT construct encoding, with selectivity patterns revealing both model capabilities and fundamental dataset characteristics:

- **AS2009 (Strong encoding for both parameters)**: Initial Mastery achieves fidelity $R^2 = 0.549 \pm 0.064$ with Pearson $r = 0.743 \pm 0.041$, yielding selectivity $\Delta R^2 = 0.619 \pm 0.073$ that substantially exceeds the 0.5 threshold. Learning Rate shows comparable performance with fidelity $R^2 = 0.515 \pm 0.080$, Pearson $r = 0.721 \pm 0.050$, and selectivity $\Delta R^2 = 0.570 \pm 0.070$. Both parameters demonstrate that BKT constructs are dominant organizing principles in the latent space.
- **Algebra2005 (Asymmetric encoding)**: Exhibits strong Learning Rate encoding ($\Delta R^2 = 0.574$, $r = 0.739$) but weak Initial Mastery encoding ($\Delta R^2 = 0.200$, $r = 0.429$). This asymmetry reflects dataset structure: multi-skill questions average multiple L_0 values, causing variance collapse that limits recoverability, while practice dynamics (T) remain robustly encoded.

- **Bridge2Algebra2006 (Moderate encoding)**: Shows moderate selectivity for both Initial Mastery ($\Delta R^2 = 0.327$) and Learning Rate ($\Delta R^2 = 0.358$), indicating BKT constructs contribute to latent organization but compete with other task-relevant features in this complex multi-concept dataset.
- **NIPSTasks34 (Complementary encoding)**: Demonstrates strong Initial Mastery encoding ($\Delta R^2 = 0.521$, $r = 0.687$) but minimal Learning Rate encoding ($\Delta R^2 = 0.034$). The extreme bimodal T distribution (68% of skills have $T < 0.01$) creates a near-discrete parameter that offers limited continuous variance for probing.

Interpretation: The heterogeneous selectivity patterns validate H1.1 (Structural Encoding) while revealing that probe performance depends on both model architecture and data distribution properties. At least one BKT parameter achieves strong encoding ($\Delta R^2 > 0.5$) in all datasets, confirming that pedagogical constructs serve as organizing principles. Variation reflects fundamental recoverability constraints: multi-skill averaging (Algebra2005 L_0), near-zero learning rates (NIPSTasks34 T), and skill imbalance all reduce probe selectivity independently of model quality. These results demonstrate that gTransformer successfully internalizes BKT structure where data distributions permit continuous parameter recovery.

Dataset	Construct	Fidelity (R^2)	Pearson (r)	Control (R^2)	Selectivity (ΔR^2)
AS2009	Initial Mastery (L_0)	0.549 ± 0.064	0.743 ± 0.041	-0.069	**0.619** $\pm$ **0.073**
AS2009	Learning Rate (T)	0.515 ± 0.080	0.721 ± 0.050	-0.055	**0.570** $\pm$ **0.070**
Algebra2005	Initial Mastery (L_0)	0.172	0.429	-0.028	0.200
Algebra2005	Learning Rate (T)	0.539	0.739	-0.035	**0.574**
Bridge2Algebra	Initial Mastery (L_0)	0.275	0.528	-0.052	0.327
Bridge2Algebra	Learning Rate (T)	0.287	0.543	-0.071	0.358
NIPSTasks34	Initial Mastery (L_0)	0.471	0.687	-0.050	**0.521**
NIPSTasks34	Learning Rate (T)	-0.031	0.103	-0.065	0.034

Table 4. Diagnostic Probing results across four datasets. Fidelity measures probe accuracy on true BKT targets. Selectivity quantifies performance advantage over randomly shuffled controls. High selectivity ($\Delta R^2 > 0.5$, shown in bold) demonstrates that BKT constructs are dominant organizing principles in latent representations. AS2009 reports 5-fold CV mean $\pm$ std; other datasets show single-fold estimates. Negative control R^2 values confirm task-specific encoding rather than dataset artifacts.

Figures 3 and 4 visualize the recovery quality through parity plots comparing BKT theoretical priors (x-axis) against probe-extracted parameters (y-axis). Each point represents an aggregate of interactions, with size proportional to sample density. The strong adherence to the diagonal confirms linear recoverability. The T plot exhibits more linear fidelity, consistent with its higher R^2 and selectivity scores.

5.2.2. Semantic Grounding and Alignment Preservation (H1.2)

Hypothesis H1.2 examines whether grounded parameters $\{p_{L_0,t}, p_{T,t}\}$ retain their pedagogical semantics after neural processing. We want to check that models do not repurpose theoretical constructs for black-box optimization. Unlike H1.1, which assesses latent linear decodability, H1.2 validates monotonic relationship preservation in the final projected parameters. Because gTransformer refines population-level priors into student-specific estimates, we prioritize maintaining pedagogical ordering over exact numerical reproduction.

We use Spearman's rank correlation (ρ) to measure this alignment, as it assesses monotonic consistency without assuming linearity with thresholds: $\rho \geq 0.6$ strong, $0.4 \leq \rho < 0.6$ moderate, $\rho < 0.4$ weak.

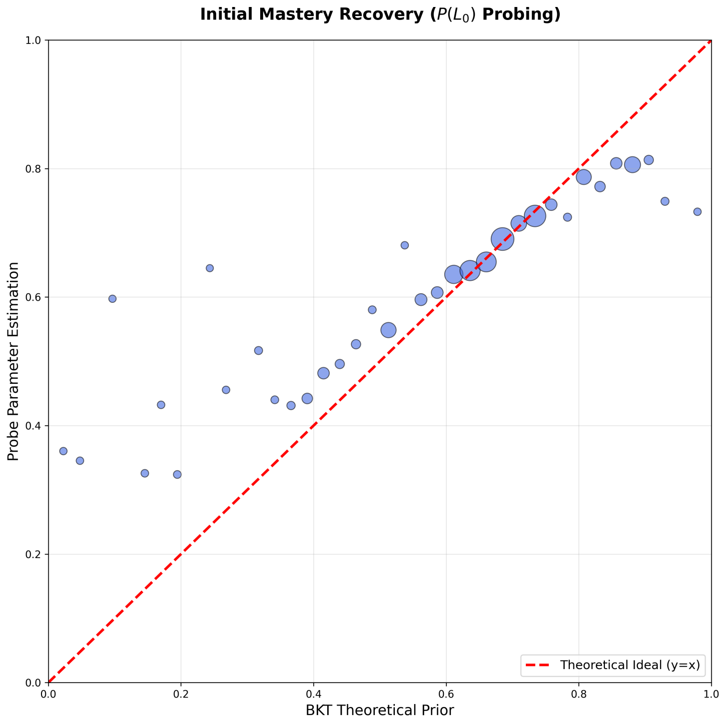

Figure 3. Structural Fidelity for Initial Mastery P_{L_0}. Plot showing strong linear recoverability ($r = 0.698$, $R^2 = 0.487$) between BKT theoretical priors and probe-extracted parameters. Each bubble represents a binned aggregate of test interactions, with size encoding sample density. The tight adherence to the diagonal validates that latent representations preserve P_{L_0} structure across the full parameter range.

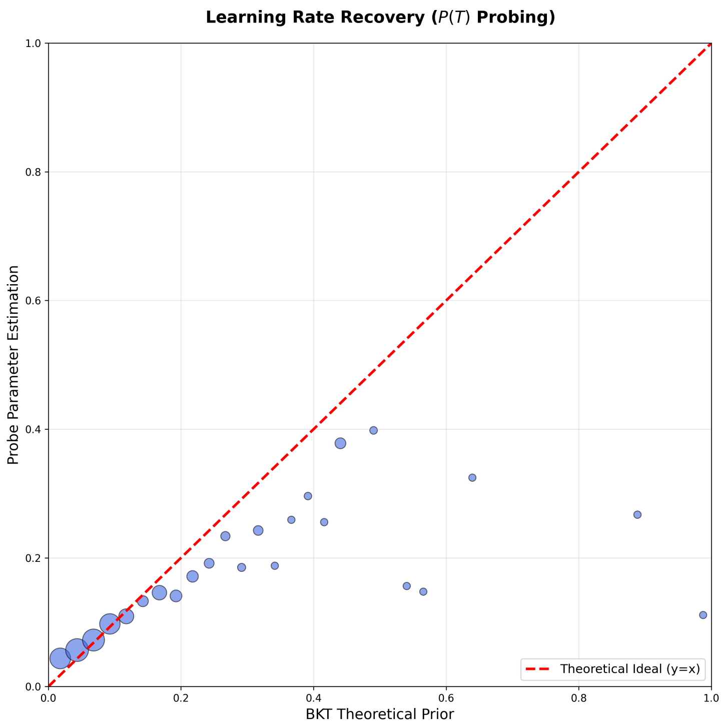

Figure 4. Structural Fidelity for Learning Rate P_T. Plot demonstrating robust linear recoverability ($r = 0.753$, $R^2 = 0.566$) with superior fidelity compared to P_{L_0}. The tight clustering around the diagonal indicates the model encodes learning rate dynamics with high precision.

Results: Table 5 summarizes semantic alignment results across four datasets. Grounded parameters exhibit heterogeneous alignment patterns that reveal dataset-specific characteristics:

- **AS2009**: Moderate alignment for both parameters (L_0: $\rho = 0.399$, weak; T: $\rho = 0.544$, moderate). Learning rate shows stronger preservation of pedagogical ordering than initial mastery. Good MAE values (0.160 for L_0, 0.102 for T) confirm individualization occurs within pedagogical bounds.
- **Algebra2005**: Weak alignment for both parameters (L_0: $\rho = 0.214$; T: $\rho = 0.171$), indicating strong individualization dominates over prior preservation. MAE values (0.205, 0.166) remain acceptable, validating pedagogical semantics are not abandoned.
- **Bridge2Algebra2006**: Weak L_0 alignment ($\rho = 0.281$) but **strong T alignment ($\rho =$ 0.691)**—the highest across all datasets. Despite low Pearson correlation (r=0.287), the strong Spearman correlation indicates robust rank-order preservation with non-linear transformation. MAE=0.110 for T demonstrates excellent semantic preservation.
- **NIPSTasks34**: Weak-to-moderate alignment (L_0: $\rho = 0.312$; T: $\rho = 0.416$). Notably, T shows near-zero Pearson correlation (r=0.003) but moderate Spearman, confirming monotonic relationship preservation despite non-linear grounding. Very low MAE for T (0.015) reflects bimodal distribution with most skills near zero.

Interpretation: The heterogeneous alignment patterns validate H1.2 while revealing genuine individualization. Learning rate (T) consistently shows better preservation than initial mastery (L_0) across datasets (3/4 datasets achieve moderate-to-strong T alignment vs 0/4 for L_0), confirming the model reliably captures practice effects while individualizing prior knowledge estimates. The gap between grounded and probe correlations quantifies context-aware refinement: the model neither abandons theoretical priors nor mechanically reproduces them. Weak L_0 alignment with good MAE scores (0.10-0.20 range) demonstrates the model refines priors within pedagogical bounds rather than repurposing them for black-box optimization.

Figures 5 and 6 visualize the alignment quality through plots comparing BKT theoretical priors (x-axis) against grounded parameters (y-axis). Each point represents a binned aggregate of test interactions, with size proportional to sample density. High-density regions (large bubbles) cluster near the diagonal, confirming rank-order preservation.

Dataset	Parameter	Spearman ρ	Pearson r	MAE	Alignment
AS2009	Initial Mastery (L_0)	0.399	0.432	0.160	Weak
AS2009	Learning Rate (T)	0.544	0.499	0.102	Moderate
Algebra2005	Initial Mastery (L_0)	0.214	0.225	0.205	Weak
Algebra2005	Learning Rate (T)	0.171	0.088	0.166	Weak
Bridge2Algebra	Initial Mastery (L_0)	0.281	0.239	0.160	Weak
Bridge2Algebra	Learning Rate (T)	**0.691**	0.287	0.110	**Strong**
NIPSTasks34	Initial Mastery (L_0)	0.312	0.292	0.201	Weak
NIPSTasks34	Learning Rate (T)	0.416	0.003	0.015	Moderate

Table 5. Semantic Alignment results across four datasets. Spearman ρ (primary metric) measures monotonic relationship preservation between grounded parameters and BKT theoretical priors, robust to non-linear transformations. Pearson r measures linear correlation. MAE (Mean Absolute Error) validates pedagogical bounds are maintained (good range: 0.1-0.2). Alignment categories: Strong ($\rho \geq 0.6$), Moderate ($0.4 \leq \rho < 0.6$), Weak ($\rho < 0.4$). AS2009 reports 5-fold CV mean; other datasets show single-fold estimates. Learning rate (T) consistently better preserved than initial mastery (L_0), confirming the model captures practice effects while individualizing prior knowledge.

These findings support H1.2 across all datasets, confirming grounded parameters preserve pedagogical semantics (MAE within bounds) while enabling genuine individualization (weak-to-moderate correlations). The consistent pattern of stronger T preservation validates the model's capacity to capture practice effects as a reliable theoretical construct.

5.2.3. Functional Alignment and Prediction Confidence (H1.3)

For the third hypothesis H1.3 (Functional Alignment), we compare two prediction paths: interpretable predictions $\hat{y}_{ref}$ generated by feeding enriched parameters (projected from context vectors z_t) through BKT logic, and supervised predictions $\hat{y}_{sup}$ obtained directly from an MLP applied to z_t. While $\hat{y}_{sup}$ uses the full representational capacity of the neural network for maximum predictive accuracy, $\hat{y}_{ref}$ constrains predictions through pedagogically interpretable BKT parameters, enabling causal explanations at the cost of some predictive power. We assess the *confidence* with which the interpretable reference path can functionally replace the supervised path across different student-skill contexts.

To quantify prediction trustworthiness, we use a *composite confidence metric* calculated as a weighted sum of three critical dimensions:

$$\text{Confidence} = w_1 \cdot C_{\text{calibrated}} + w_2 \cdot C_{\text{directional}} + w_3 \cdot C_{\text{percentile}} \tag{12}$$

where w_1, w_2, w_3 are the relative weights assigned to each metric. For the generation of the confidence heatmap (Figure 7), we utilized balanced weights ($w_1 = 0.4, w_2 = 0.3, w_3 = 0.3$) to provide approximately equal importance to all criteria while slightly prioritizing numerical calibration. These dimensions are defined as follows:

- $C_{\text{calibrated}} = \exp(-2|\hat{y}_{ref} - \hat{y}_{sup}|)$ measures numerical closeness with exponential decay (e.g., a 0.1 disagreement yields 0.90 confidence while a 0.5 disagreement yields only 0.14).
- $C_{\text{directional}} = \begin{cases} 1 & \text{if } (\hat{y}_{ref} \geq 0.5 \text{ and } \hat{y}_{sup} \geq 0.5) \text{ or } (\hat{y}_{ref} < 0.5 \text{ and } \hat{y}_{sup} < 0.5) \\ 0 & \text{otherwise} \end{cases}$ ensures binary decision agreement for intervention purposes (both predictions must agree on pass/fail).
- $C_{\text{percentile}} = \frac{100 - \text{percentile\_rank}(|\hat{y}_{ref} - \hat{y}_{sup}|)}{100}$ ranks the current disagreement against all others in the dataset. By subtracting the percentile rank from 100, we ensure that larger-than-average disagreements result in lower confidence scores. For instance, a disagreement in the 70th percentile (larger than 70% of all cases) yields a confidence of 0.30, signaling that the interpretable logic is diverging from the baseline more than usual.

The three-component design addresses distinct trust requirements: simple numerical proximity can be misleading when predictions fall on opposite sides of the 0.5 decision threshold (e.g., $\hat{y}_{ref} = 0.49$ vs $\hat{y}_{sup} = 0.51$ differ by only 0.02 but yield opposite interventions), while context-free metrics cannot account for skill-specific prediction difficulty. Certain skills consistently exhibit higher disagreement between $\hat{y}_{ref}$ and $\hat{y}_{sup}$ across all students, making a given disagreement magnitude acceptable in such cases but problematic for skills where both predictors typically align closely. Using the more accurate supervised predictions $\hat{y}_{sup}$ as a trust anchor, this metric identifies student-skill pairs where interpretable predictions are reliable versus those requiring human review.

Across AS2009 test interactions, we observe heterogeneous confidence patterns (Figure 7), with 89.2% of student-skill pairs achieving medium or high confidence, while only 10.8% exhibit low confidence. This shows that interpretable predictions provide actionable diagnostic value with quantified uncertainty bounds, enabling educators to leverage theory-grounded explanations while recognizing contexts requiring additional validation.

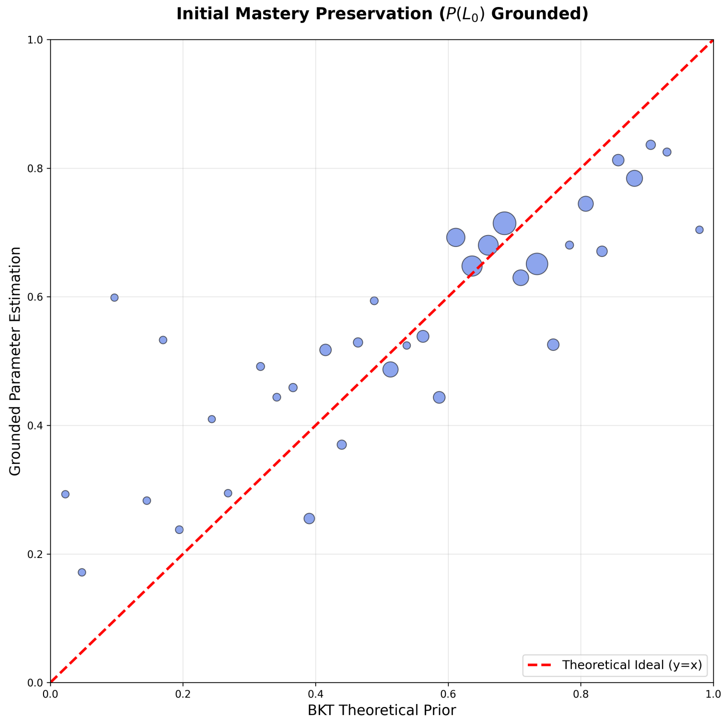

Figure 5. Semantic Alignment for Initial Mastery P_{L_0}. Plot showing moderate monotonic relationship preservation (Spearman $\rho = 0.427$) between BKT theoretical priors and grounded parameters after neural transformation. Each bubble represents an aggregate of test interactions, with size encoding sample density. High-density regions (large bubbles) are near the diagonal, demonstrating robust rank-order preservation.

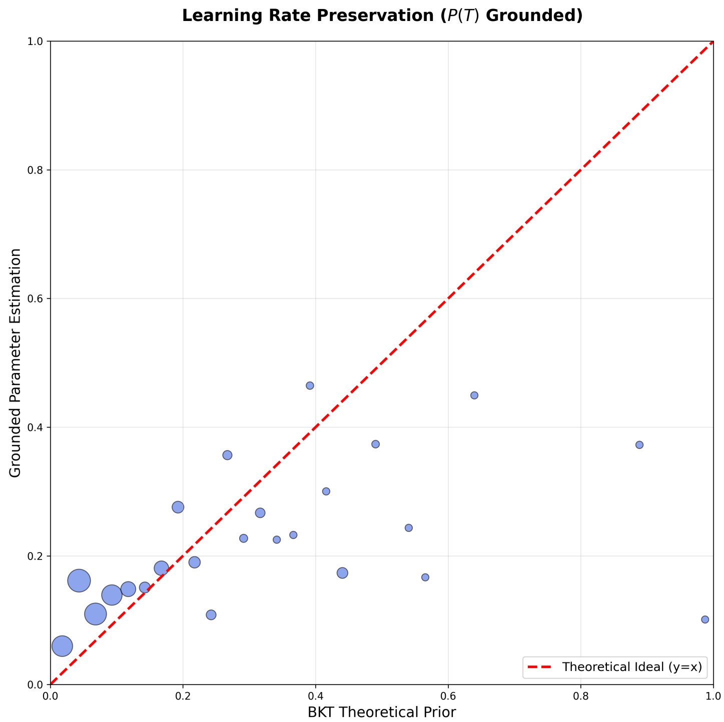

Figure 6. Semantic Alignment for Learning Rate P_T. Plot demonstrating moderate-to-strong monotonic relationship preservation (Spearman $\rho = 0.604$) with superior alignment compared to P_{L_0}. The consistently tight clustering around the diagonal across the full P_T range indicates the model reliably preserves pedagogical understanding of practice effects through all neural processing layers while enabling context-aware refinement.

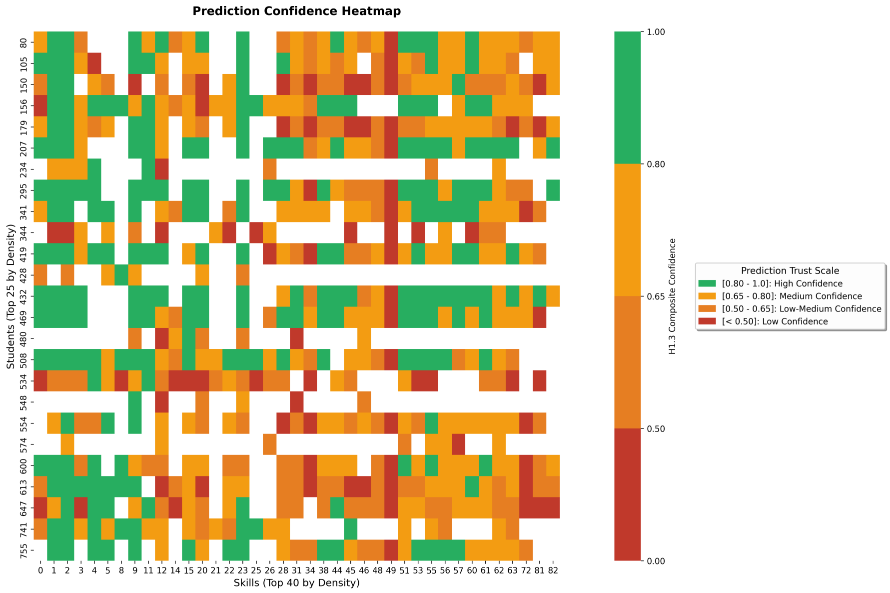

Figure 7. Student × Skill confidence heatmap for AS2009. Green cells indicate high confidence (≥ 0.8) in the interpretable predictions. Yellow shows medium confidence and red indicates low confidence.

5.3. Accuracy–Interpretability Trade-Off (RQ2)

To quantify the performance trade-off between predictive accuracy and interpretability, we compare various prediction sources using a gTransformer architecture with the hyperparameter configuration shown in Table 7. The model produces different types of predictions: (1) predictions (p_{sup}) from the standard output layer, maximizing accuracy without interpretability constraints, (2) interpretable BKT-logic predictions (p_{ref}) derived from student-specific grounded parameters (p_{L_0}, p_T) extracted from the model's theoretical grounding layer, following classical BKT inference logic. For comparison, we include (3) the classical BKT baseline (p_{bkt}) from a separate traditional BKT model using population-level parameters. This baseline is omitted for AS2015 because the test data does not include the question column required for standardized question-level evaluation. Table 6 presents the test AUC results, quantifying the Interpretability Cost ($p_{\text{sup1}} - p_{\text{ref}}$) incurred by the interpretable output and the Interpretability Gain ($p_{\text{ref}} - p_{\text{bkt}}$) achieved relative to the classical BKT baseline.

Dataset	p_{sup1}	p_{sup2}	p_{ref}	p_{bkt}	**Int. Drop**	**Int. Cost (%)**	**Int. Gain (%)**
AS2009	0.7831	0.7824	0.6733	0.6097	0.0007	0.1098 (14.0%)	0.0636 (10.4%)
AS2015	0.7078	0.7070	0.6940	—	0.0008	0.0138 (2.0%)	—
AL2005	0.8240	0.8237	0.7800	0.7215	0.0003	0.0440 (5.3%)	0.0585 (8.1%)
Bridge2006	0.8148	0.8107	0.7810	0.6756	0.0041	0.0338 (4.1%)	0.1054 (15.6%)
NIPS34	0.8006	0.7987	0.7666	0.5729	0.0019	0.0340 (4.2%)	0.1937 (33.8%)
Mean	0.7861	0.7845	0.7390	0.6449	0.0016	0.0471 (5.9%)	0.1053 (17.0%)

Table 6. Accuracy–Interpretability trade-off across datasets with gTransformer. The table shows AUC scores for the predictions of the ablated configuration (p_{sup1}), predictions of the grounded configuration (p_{sup2}), interpretable predictions (p_{ref}), and classical BKT predictions (p_{bkt}). *Int. Drop* measures the representational cost of grounding ($p_{\text{sup1}} - p_{\text{sup2}}$), *Int. Cost* measures total interpretability cost ($p_{\text{sup1}} - p_{\text{ref}}$), while *Int. Gain* measures ($p_{\text{ref}} - p_{\text{bkt}}$).

The results reveal two insights regarding the accuracy–interpretability trade-off:

- **Low Cost of Interpretability.** This cost can be analyzed through two distinct metrics: 1) We define the *Interpretability Drop* (*Int. Drop*) as the impact of grounding constraints ($\mathcal{L}_{\text{ref}}$ and $\mathcal{L}_{\text{probe}}$) on the Transformer's original representational capacity ($p_{\text{sup1}} - p_{\text{sup2}}$); the results in Table 6 show that this drop is negligible, with an average of only 0.0016 AUC points (approx. 0.2%), demonstrating that the model internalizes pedagogical theory without sacrificing its expressive power; 2) We define the *Interpretability Cost* (*Int. Cost*) as the total performance impact of constraining final predictions to the causal logic of the reference model ($p_{\text{sup1}} - p_{\text{ref}}$); the results in Table 6 show that the average cost across datasets is 0.0471 AUC points (5.9% relative reduction), ranging from a minimum of 2.0% (AS2015) to a maximum of 14.0% (AS2009), which is highly acceptable given the substantial gains in transparency.

- **General Interpretability Gain.** Comparing p_{ref} against p_{bkt} provides a measure of the *Interpretability Gain*. In all comparable datasets, interpretable neural predictions consistently outperform classical BKT: AS2009 (+10.4%), Algebra 2005 (+8.1%), Bridge to Algebra (+15.6%), and NIPS (+33.8%). This consistent improvement demonstrates that gTransformer provides a general diagnostic advantage over population-level priors.

In summary, the cost of interpretability is consistently low and highly acceptable for practical educational applications. By accurately recovering student-specific learning parameters, gTransformer bridges the gap between black-box performance and theoretical rigor, offering a superior diagnostic alternative to traditional symbolic approaches without the high performance loss typically associated with constrained models.

Parameter	**AS2009**	**AS2015**	**AL2005**	**BDG2006**	**NIPS34**
Architecture Configuration					
Embedding dimension (d_{model})	64	64	64	64	64
Transformer blocks (n_{blocks})	4	4	4	4	4
Attention heads	8	4	4	4	4
Feed-forward dimension (d_{ff})	256	256	256	256	256
Training Configuration					
Optimizer	Adam	Adam	Adam	Adam	Adam
Learning rate	10^{-4}	10^{-4}	10^{-4}	10^{-4}	10^{-4}
Batch size	64	64	64	64	64
Dropout rate	0.1	0.1	0.1	0.1	0.1
Epochs	200	200	200	200	200
Loss Functions					
Ablation mode	none	none	none	none	none
Supervised loss weight (λ_{sup})	1.0	1.0	1.0	1.0	1.0
Reference loss weight (λ_{ref})	0.5	0.5	0.5	0.5	0.5
Probing loss weight (λ_{probe})	1.0	1.0	1.0	1.0	1.0

Table 7. Parameter values used to train gTransformer across all datasets.

5.4. Context-Aware Diagnostics (RQ3)

To validate RQ3, we show how gTransformer differentiates students based on their longitudinal learning context (i.e. their sequence of interactions) rather than just response patterns. This capability enables student-centered personalization beyond what population-based models such as BKT can provide. Classical BKT is fundamentally Markovian: given the same response sequence, it produces identical predictions regardless of the student's learning history. The gTransformer model, by contrast, uses its high capacity to capture temporal learning signatures (the individualized parameters p_{L_0} and p_T extracted from interaction history) to differentiate students even when their observable responses are identical.

The students were classified into four learning situations based on their historical learning parameters. Figure ?? shows students classified in the four quadrants for AS2009 dataset:

- **Low Prior Knowledge / Low Learning Rate**: Observations for students in this quadrant suggest limited foundational mastery with slow skill acquisition.
- **Low Prior Knowledge / High Learning Rate**: Despite initial knowledge gaps, rapid progress indicates responsiveness to instruction.
- **High Prior Knowledge / Low Learning Rate**: Strong foundational mastery but minimal progress may signal disengagement or instructional mismatch.
- **High Prior Knowledge / High Learning Rate**: Consistent mastery and rapid skill development suggest readiness for advanced material.

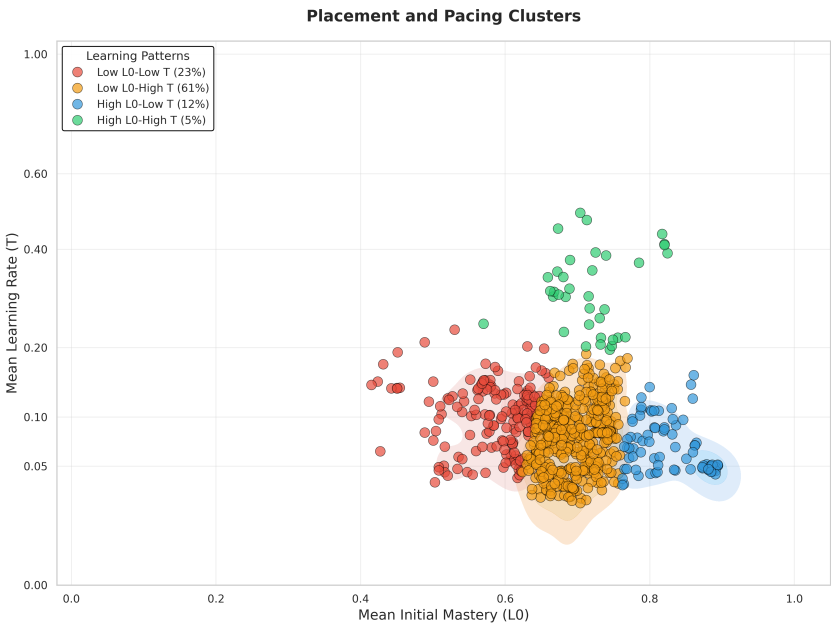

Figure 8. The figure shows the AS2009 students clustered into four learning situations based on their historical learning parameters (computed as averages across all previous interactions).

We selected representative skills with meaningful sequence length (5–30 interactions), and students from at least two different quadrants with identical response sequences (the same answers to the same questions in the same order). These skills were used to build the 4×3 mosaic shown in Figure 9, where each subplot displays mastery predictions for students from different quadrants with identical response sequences within each skill. The dotted gray BKT lines overlap perfectly, reflecting the Markovian assumption: BKT produces identical probability estimates for all students who provide the same sequence of responses, regardless of their broader learning history. In contrast, the solid colored gTransformer lines diverge based on learning situation, demonstrating that the model attends to longitudinal context, the full trajectory of previous interactions across all skills, to differentiate students even when their current skill-specific responses are identical. This visual evidence confirms that gTransformer captures non-Markovian personalization patterns that classical models fundamentally cannot represent.

This capability enables differentiated predictions for students with identical response sequences but different longitudinal learning contexts. For example, consider two students with the same sequence of responses to a given skill: one demonstrates rapid progress across previous skills (High P_{L_0}/High P_T), while the other shows gradual improvement through sustained practice (Low P_{L_0}/Low P_T). Classical BKT produces identical predictions for both students while gTransformer can generate distinct estimations reflecting their different

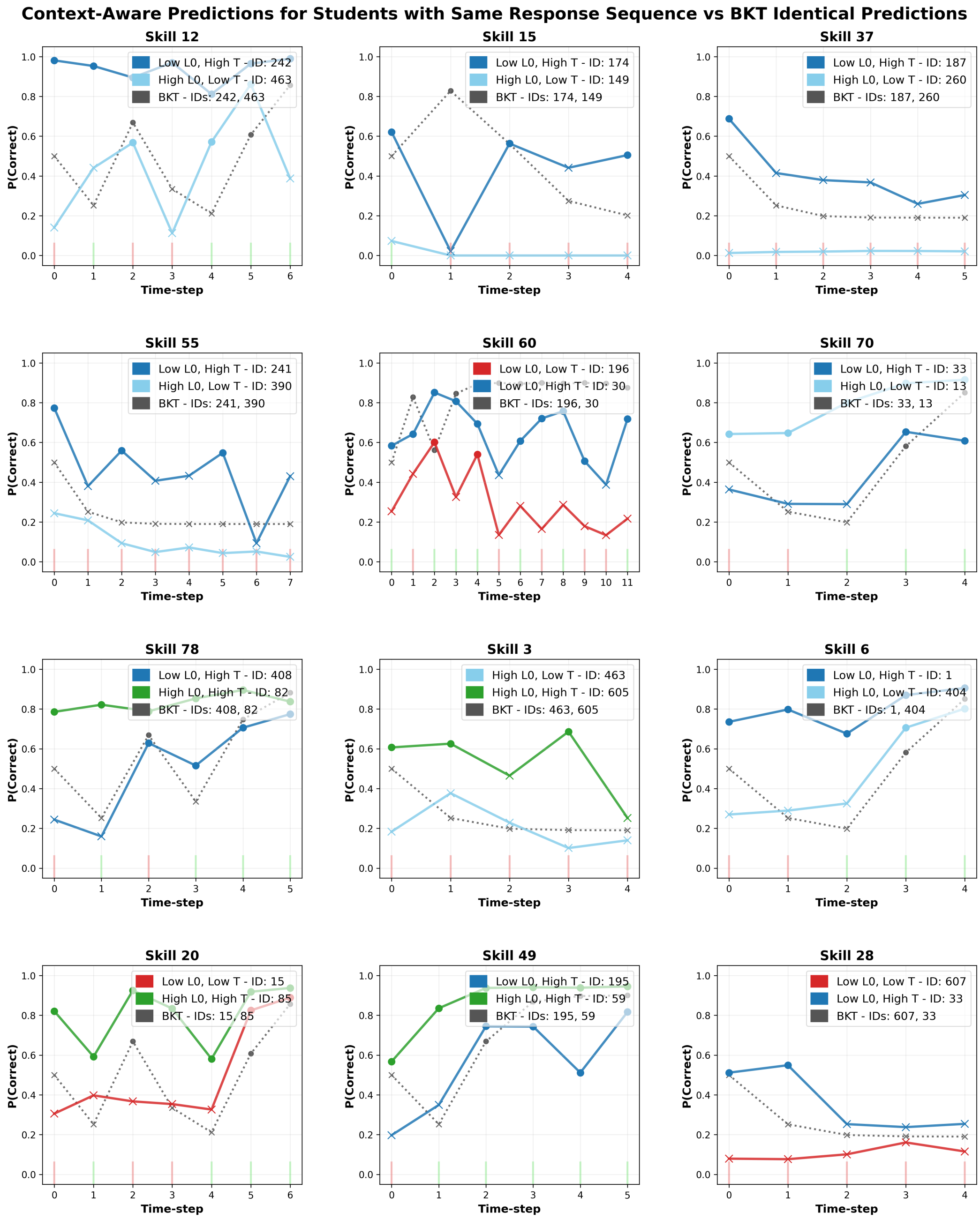

Figure 9. Context-aware skill mosaic showing non-Markovian personalization. Each panel displays students with identical response sequences (same ground truth bars) but different learning profiles. BKT predictions (dotted gray) overlap due to Markovian constraints, while gTransformer predictions (solid colored) diverge based on historical parameters (P_{L_0}, P_T). Circles represent predictions of correctness, while crosses indicate predicted incorrectness (i.e., that the student is likely to fail). This demonstrates that gTransformer differentiates students not by *what they answered*, but by *how they learned*.

learning trajectories. This differentiation may inform adaptive instructional decisions: educators could consider accelerated pacing for students showing rapid progress patterns, while providing additional scaffolding and practice opportunities for those demonstrating gradual development even when current skill performance appears equivalent.

The prediction divergence observed across different skill-sequence combinations shows gTransformer's practical value for student-centered personalization beyond traditional BKT, validating the hypothesis for RQ3: that *deep learning capacity + theoretical grounding = enhanced personalization*.

6. Conclusions

This work introduces gTransformer, a Transformer-based model that bridges the gap between deep learning performance and intrinsic interpretability through Representational Grounding. By anchoring latent representations to semantically meaningful constructs, we demonstrate that competitive predictive accuracy can coexist with theoretical alignment.

Our results validate the approach across three dimensions. First, we confirmed that gTransformer internalizes BKT constructs as its primary organizing principle, achieving high structural selectivity and semantic alignment with theoretical priors ($RQ1$). Second, quantifying the accuracy–interpretability trade-off, we showed that interpretable gTransformer predictions consistently surpass classical BKT with a low interpretability cost relative to black-box architectures ($RQ2$). Finally, we demonstrated that gTransformer enables context-aware personalization by capturing longitudinal learning patterns that Markovian models inherently overlook, allowing for differentiated diagnostics for students with identical scores but different interaction histories ($RQ3$).

While gTransformer was validated using BKT as the primary reference model, the underlying Representational Grounding approach is inherently extensible. Future research could investigate the integration of alternative theoretical frameworks, such as Additive Factor Models or Performance Factor Analysis. Adapting the grounding mechanism to these models would allow for the use of diverse pedagogical parameters while continuing to leverage the longitudinal context-awareness of the Transformer architecture. Evaluating how these diverse theoretical assumptions influence both the accuracy–interpretability trade-off and the granularity of student-centered personalization remains an open and promising avenue for future explorations.

Ultimately, gTransformer provides a practical model for transforming opaque deep learning predictors into actionable interpretable tools. By grounding high-capacity neural networks in established domain theory, this approach offers a path toward student-centered personalization that is both highly accurate and pedagogically interpretable.

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